

Technical Specification and Radio Compliance of the Motion CAM-RF System

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1 Introduction

This document presents the design and technical specifications of a presence-detection system based on a PIR sensor and an ESP32-CAM camera module. When motion is detected, a M5Stack Tab5 tablet automatically displays the video stream transmitted by the camera over a Wi-Fi connection.

The purpose of this report is to provide a comprehensive and structured description of the system's operation, as well as its functional, hardware, and radio-frequency requirements, in order to facilitate its understanding, development, testing, and future evolution.

A GitHub repository has been created to centralize all project-related resources:

<https://github.com/BriceBen/MOTION-CAM-RF.git>

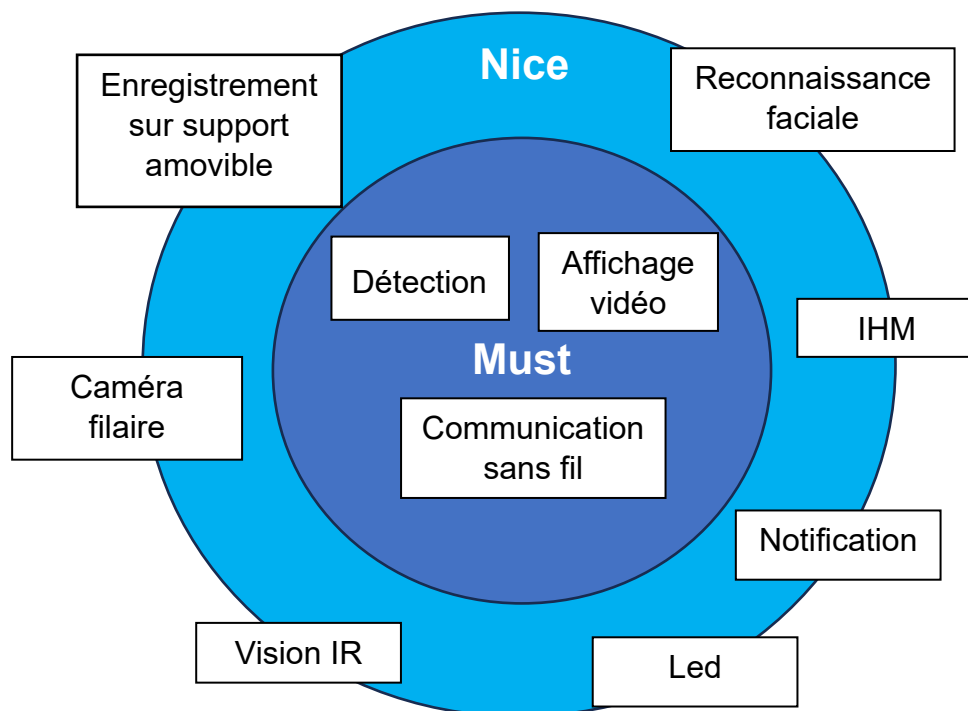


Diagramme des exigences Must/Nice

2 Functional Analysis

The purpose of the functional analysis is to describe the Motion CAM-RF system through the functions it must perform and the constraints that arise from them.

It clearly identifies what the system is expected to do (its main and technical functions) and under which conditions it must operate (its constraints and performance requirements).

This approach, derived from system functional analysis methodologies, aims to establish a logical link between the expressed needs and the technical solutions that have been implemented.

This section illustrates how the various hardware components and network elements interact with one another, explaining the role of each module within the overall system architecture.

2.1 Block Diagram

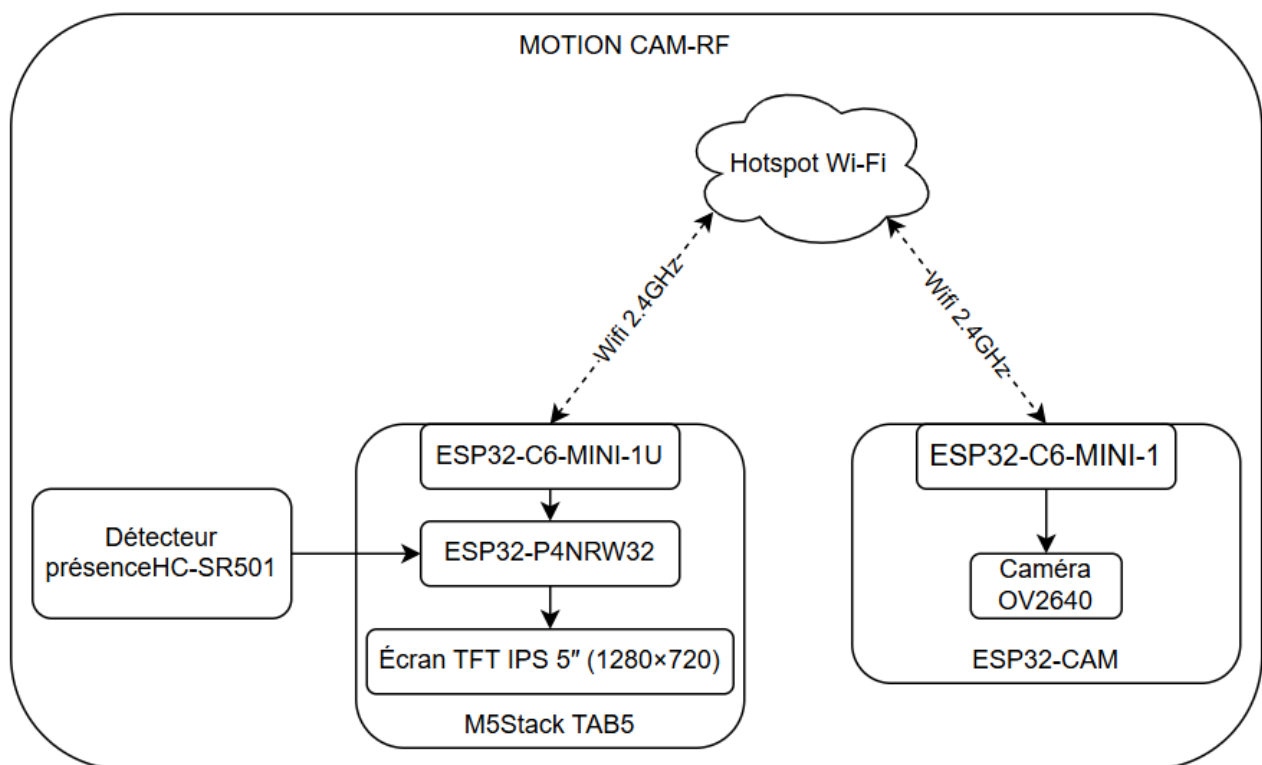


Schéma de principe MOTION CAM-RF

This diagram illustrates the overall architecture of the system and the interconnection of its different subsystems. Both ESP32 modules (the Tab5 and the camera) are connected to the same local network through the 2.4 GHz Wi-Fi hotspot. The HC-SR501 motion sensor detects presence and sends a logic signal to the M5Stack Tab5, which centralizes the event processing.

In parallel, the ESP32-CAM handles video capture and streaming over the 2.4 GHz Wi-Fi network provided by the smartphone acting as an access point. When motion is detected, the Tab5 sends a request to the camera to display the live video stream on its screen.

2.2 Main Function

FP1: Motion Detection and Real-Time Video Display

The primary function of the Motion CAM-RF system is to detect the presence of motion and automatically display the corresponding video stream.

More specifically, the HC-SR501 passive infrared sensor detects thermal variations within its field of view. As soon as motion is detected, it sends a logic-level signal to the M5Stack Tab5, which acts as the central control unit.

The Tab5 then establishes communication over the 2.4 GHz Wi-Fi network with the ESP32-CAM in order to open and display the live video stream on its screen.

This main function relies on the Wi-Fi radio link, which forms the core of the system. It enables fast and reliable wireless transmission of the MJPEG stream between the ESP32 modules, ensuring near-instantaneous reaction (latency < 5 s).

The quality of this communication is essential to the system's performance, as it directly affects the continuity of the video stream and the stability of the display.

2.3 Constraint Functions

Beyond its main function, the system must comply with a set of technical, environmental, and regulatory constraints:

FC1: Ensure a display latency of less than 5 seconds

- The system must display the video stream on the Tab5 in under five seconds after motion is detected by the PIR sensor. This constraint guarantees a sufficiently responsive operation for near-real-time monitoring.

FC2: Maintain stable communication over a range of up to 10 meters

- The Wi-Fi connection must remain stable in an indoor environment at distances of up to 10 meters, even in the presence of light or thick walls.
- FC3: Power all modules with a 5-volt supply
- All components of the system (PIR sensor, Tab5, and ESP32-CAM) must be powered at 5 V from a common source without a built-in battery, ensuring electrical compatibility and simplifying the overall assembly.

FC4: Ensure compatibility with the smartphone's 2.4 GHz Wi-Fi network

- The system must automatically connect to the smartphone hotspot operating in WPA2 on the 2.4 GHz band. This constraint ensures interoperability between the two ESP32 modules and maintains stable radio communication.

FC5: Limit transmission power to 20 dBm (100 mW EIRP)

- The Wi-Fi transmission power must remain less than or equal to 20 dBm, in accordance with regulatory limits in the 2.4 GHz ISM band. No additional amplifiers or antenna modifications may be added.

FC6: Guarantee safe indoor operation

- The system is intended for indoor use within a temperature range of 10 °C to 40 °C. This constraint helps prevent malfunctions related to temperature variations or humidity.

2.4 Technical Functions

The technical functions translate the functional requirements into concrete technical solutions. They specify the hardware and software means implemented to achieve the main function (FP1) while complying with all constraint functions.

The technical functions describe the hardware and software mechanisms used to deliver the primary purpose of the Motion CAM-RF system, while ensuring that performance requirements and radio-frequency compliance constraints are fully met.

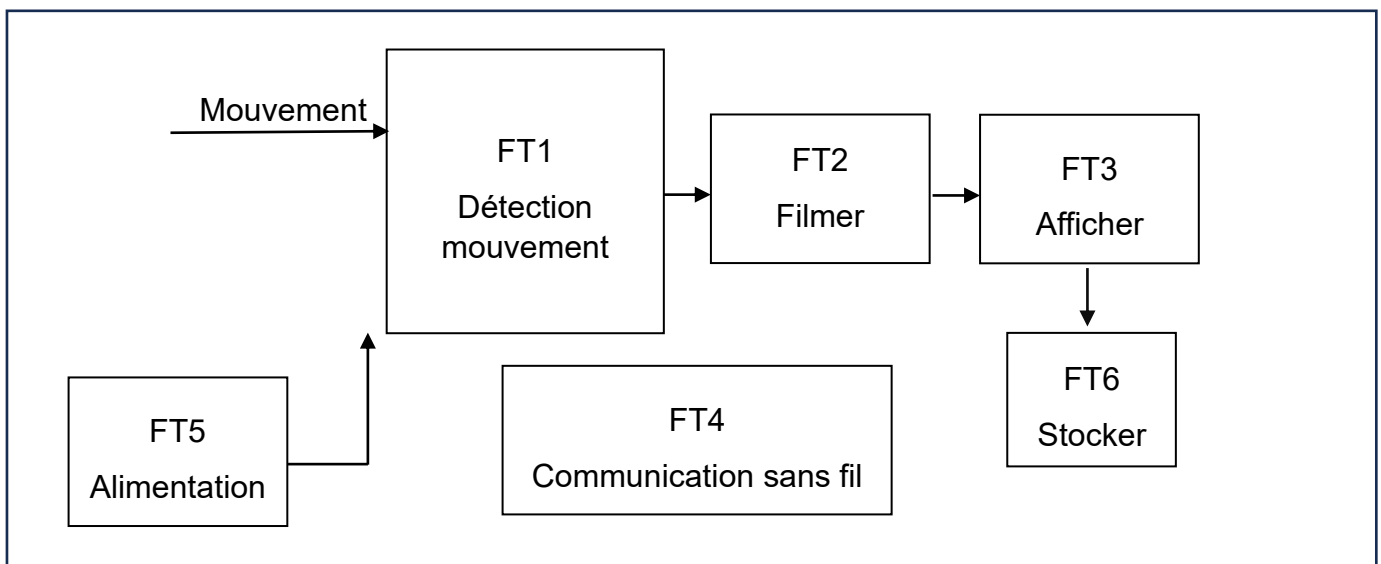


Schéma des fonctions techniques

FT1: Detect Motion

- The HC-SR501 passive infrared sensor detects thermal variations within its field of view and sends a logic signal to the M5Stack Tab5 through a GPIO input.
- The software applies a 500 ms debounce filter and a 30-second hold time to avoid unwanted repeated triggers.
- This detection event initiates the entire operating sequence of the system.

FT2: Capture the Detected Scene

- The ESP32-CAM captures the video stream in real time as soon as it receives a request from the Tab5.
- It encodes the images in MJPEG format and streams them through a local HTTP server.
- Image quality, resolution, and frame rate can be adjusted in the firmware depending on network conditions and wireless range.

FT3: Display the Video Stream

- The M5Stack Tab5 connects to the camera's IP address and displays the stream in full screen on its graphical interface.
- The display must start within 5 seconds after detection and remain active for 30 seconds after the last trigger before automatically turning off.

FT4: Configure and Maintain the Wireless Link

- The ESP32-CAM and Tab5 communicate through the smartphone's 2.4 GHz Wi-Fi hotspot (WPA2).
- The communication must remain stable up to a distance of 10 meters, with a transmission power ≤ 20 dBm (100 mW EIRP), in accordance with the RED 2014/53/EU directive and the EN 300 328 and EN 301 489-17 standards.
- The system must automatically manage connection and reconnection in case of link loss, without requiring user intervention.

FT5: Power Supply

- All modules (PIR sensor, Tab5, ESP32-CAM) are powered with a +5 V supply.

FT6: Store Data

- Images must be saved onto removable storage.

3 Requirements

A summary file titled “RADIO ELECTRONIC SYSTEM SPECIFICATIONS” (generated using the claude.ai form) is provided in the appendix.

3.1 Radio (EN300328)

This table presents the main requirements of the ETSI EN 300 328 V2.2.2 (2019-07) standard applicable to the Motion CAM-RF system, which operates over 2.4 GHz Wi-Fi (IEEE 802.11 b/g/n). The limits and criteria listed below are extracted from the normative clauses (§ 4.3.2.x) and the corresponding test methods (§ 5.4.x), adapted to a wideband non-FHSS device.

Exigence	Limite / Critère	Clause ETSI & Méthode d'essai
Bande de service	2400 – 2483,5 MHz	§ 4.3.1 / Table 1
Puissance RF (EIRP)	≤ 20 dBm (100 mW)	§ 4.3.2.2 / Essai 5.4.2
Densité de puissance (PSD)	≤ 10 dBm/MHz EIRP	§ 4.3.2.3 / Essai 5.4.3
Duty cycle / Séquence TX	Tx-sequence ≤ 10 ms ; Tx-gap ≥ 3,5 ms ; duty ≤ valeur déclarée	§ 4.3.2.4 / 5.4.2
Accès au canal (Adaptivity / LBT-CSMA)	Respect du mécanisme LBT/CSMA ; TxOn/(TxOn + TxOff) ≤ 10 % / 50 ms	§ 4.3.2.6 / 5.4.6
Largeur de canal occupée (OCBW)	99 % de la puissance dans la bande ; OCBW ≤ 20 MHz si P > 10 dBm	§ 4.3.2.7 / 5.4.7
Émissions hors bande (OOB)	Conformité au masque spectral ETSI	§ 4.3.2.8 / 5.4.8
Émissions parasites TX	≤ -36 dBm (30 MHz–1 GHz) ; ≤ -30 dBm (1–12,75 GHz)	§ 4.3.2.9 + Table 12 / 5.4.9
Émissions parasites RX	≤ -57 dBm (30 MHz–1 GHz) ; ≤ -47 dBm (1–12,75 GHz)	§ 4.3.2.10 + Table 13 / 5.4.10
Sensibilité du récepteur	PER ≤ 10 % pour niveau ≥ -80 dBm (minimum)	§ 4.3.2.11 / Critère A
Sélectivité co-canal / adjacente	Maintien du service avec signal utile – brouilleur ≤ 3 dB / 15 dB	§ 4.3.2.11 Tables 14–15
Blocking du récepteur	Pas de perte de service pour brouilleur offset ± 10–30 MHz (-34 à -10 dBm)	§ 4.3.2.11 Table 16 / 5.4.11
Rejection hors-bande RX	Aucune dégradation lors d'un brouillage hors bande (> 1 GHz)	§ 4.3.2.10 / Annex B
Disponibilité du canal (CCA)	Temps moyen d'accès < 10 ms ; accès ≥ 90 % (802.11 CSMA/CA)	§ 4.3.2.6 / Annex E
Conditions d'essai environnementales	15–35 °C ; 25–75 % HR ; tension nominale 5 V	§ 5.2

3.2 Cybersecurity (EN303645)

This table presents the cybersecurity requirements applicable to the Motion CAM-RF project, in accordance with the RED Directive 2014/53/EU (Articles 3(3)(d/e/f)) and Delegated Regulation (EU) 2022/30. These requirements will be covered by the harmonised standards EN 18031-1, EN 18031-2, and EN 18031-3, which will become mandatory from 1 August 2025 for all connected radio equipment.

Exigence	Limite / Critère	Référence & Méthode d'essai
Protection du réseau et de son fonctionnement	L'équipement ne doit pas perturber le fonctionnement du réseau Wi-Fi ni abuser de ses ressources. Implémentation d'un filtrage MAC/IP et limitation des connexions.	RED Art. 3(3)(d) – EN 18031-1 §4.3
Résilience logicielle et mises à jour	Le firmware doit être signé numériquement et vérifiable. Les mises à jour OTA doivent être sécurisées (TLS / authentification).	EN 18031-1 §4.3 / IEC 62443-4-1
Authentification et contrôle d'accès	Les interfaces Wi-Fi et HTTP doivent être protégées par mot de passe fort (≥ 12 caractères) et protocole WPA2-PSK.	EN 18031-1 §5.1
Protection des données et vie privée	Les flux vidéo ne doivent pas quitter le réseau local sans consentement. Aucune transmission externe non sécurisée.	RED Art. 3(3)(e) – EN 18031-2 §6.2
Chiffrement et intégrité des communications	Utilisation du chiffrement WPA2-AES ; possibilité d'évolution vers TLS 1.2+ pour les communications externes.	EN 18031-2 / ISO 27001 / NIST SP 800-57
Protection contre la fraude et le détournement de fonction	Aucune commande non authentifiée ne doit pouvoir détourner le flux vidéo ou exécuter du code externe.	RED Art. 3(3)(f) – EN 18031-3 §5.1
Gestion des accès physiques et logiques	Les ports de debug (UART/USB) doivent être désactivés ou sécurisés en utilisation normale.	EN 18031-1 §6 / EN 62443-4-2
Gestion des vulnérabilités et maintenance	Suivi des vulnérabilités (CVE, Espressif advisories) et mise à jour des correctifs de sécurité.	EN 18031-1 Annex A / ISO 30111
Conformité réglementaire	Les exigences RED 3(3)d/e/f deviennent obligatoires à partir du 01/08/2025 pour les équipements connectés.	Règlement (UE) 2022/30

3.3 Health & Safety (EN 62368-1 / EN 62479)

This table presents the main health and safety requirements applicable to the Motion CAM-RF equipment, in accordance with the RED Directive 2014/53/EU (Article 3(1)(a)) and the harmonised standards EN 62368-1 and EN 62479.

Exigence	Limite / Critère d'acceptation	Référence normative & Méthode d'essai
Protection contre les chocs électriques	Les parties accessibles doivent être à très basse tension (≤ 60 V DC). Fonctionnement sous 5 V.	EN 62368-1 §5.2 / §5.4
Protection thermique	Température de surface ≤ 70 °C pour les surfaces manipulables.	EN 62368-1 §7.4
Stabilité et montage	Résistance aux contraintes mécaniques normales (chute, vibrations légères).	EN 62368-1 §4.1 & Annexe B
Protection contre le feu / matériaux	Matériaux plastiques auto-extinguibles (UL94 V-1 ou mieux).	EN 62368-1 §9.3
Sécurité de l'alimentation	Protection contre surintensités ($>150\%$). Ports USB-C limités à 5 V.	EN 62368-1 §5.4.4
Exposition RF (champ électromagnétique)	Puissance ≤ 100 mW EIRP, distance ≥ 20 cm $\rightarrow$ pas de mesure SAR requise.	EN 62479 §4 / Annexe B
Rayonnement non ionisant (RF)	Densité de puissance ≤ 10 W/m ² pour 2,4 GHz ; conforme ICNIRP.	EN 62479 §4.2
Utilisation prévue	Usage intérieur, 10–40 °C, pas de contact prolongé du corps.	EN 62368-1 §1.4 / EN 62479 §3

3.4 EMC (EN301489-1 / EN301489-17)

This table summarises the main electromagnetic compatibility requirements for the Motion CAM-RF equipment, in accordance with the RED Directive 2014/53/EU (Article 3(1)(b)) and the EN 301 489-1 and EN 301 489-17 standards.

Exigence	Limite / Critère d'acceptation	Référence normative & Méthode d'essai
Emissions conduites	≤ 46 dB μ V (150 kHz–30 MHz) sur alimentation DC.	EN 301 489-1 §5.3 / CISPR 32
Emissions rayonnées	≤ 40 dB μ V/m (30–230 MHz) ; ≤ 47 dB μ V/m (230–1000 MHz) à 10 m.	EN 301 489-1 Annexe B / CISPR 32
Immunité électrostatique (ESD)	± 4 kV contact, ± 8 kV air : aucune perte de fonction.	EN 301 489-1 §5.4.1 / IEC 61000-4-2
Immunité aux transitoires rapides (EFT/Burst)	± 1 kV sur lignes DC : aucune perte de fonction.	EN 301 489-1 §5.4.2 / IEC 61000-4-4
Immunité aux surtensions (Surge)	$\pm 0,5$ kV sur ports d'alimentation : fonctionnement maintenu.	EN 301 489-1 §5.4.3 / IEC 61000-4-5
Immunité champ RF rayonné	3 V/m (80 MHz–2,7 GHz) : pas de perturbation Wi-Fi.	EN 301 489-17 §5.5 / IEC 61000-4-3
Immunité champ conduit RF	3 V (150 kHz–80 MHz) : communication maintenue.	EN 301 489-1 §5.4.4 / IEC 61000-4-6
Critère de performance A/B	A : fonctionnement normal ; B : reprise automatique sans intervention.	EN 301 489-1 §4.2

4 Response to Requirements

Élément	Rôle	Connexion	Paramètres	Référence module
Smartphone	AP/Hotspot	2,4 GHz, WPA2	Canal Auto	Inconnu
ESP32-CAM	Caméra IP	Client Wi-Fi	HTTP MJPEG	ESP32-C6-MINI-1
M5Stack Tab5	Afficheur/Contrôleur	Client Wi-Fi	HTTP client	ESP32-C6-MINI-1

Équipement du MOTION-CAM RF

4.1 Standards and FCC

As part of the regulatory assessment of the Motion CAM-RF system, a comparative analysis was carried out between the compliance tests performed on the ESP32-C6-MINI-1 radio module (FCC report, ID: 2AC7Z-ESPC6MINI1, available in the GitHub repository) and the requirements of the European standards applicable under the RED Directive 2014/53/EU.

The tests conducted in accordance with FCC §15.247 cover all parameters required by EN 300 328, including transmit power, occupied bandwidth, and out-of-band emissions.

The approach adopted consists of comparing the measurement methods, regulatory limits, and verified parameters in the FCC report with those defined in the corresponding European harmonised standards: EN 300 328, EN 301 489-1/-17, EN 62368-1, EN 62311, and EN 62479.

This analysis is presented in the form of a correspondence table between U.S. and European requirements, followed by an interpretation to conclude on the module's level of compliance with the European regulatory framework.

Domaine	Norme CE correspondante	Équivalence ou compatibilité FCC
Radio (RF)	EN 300 328 V2.2.2	Le rapport FCC cite Part 15 Subpart C §15.247, couvrant la bande ISM 2,4 GHz (2400–2483,5 MHz). Les essais de bande passante, densité spectrale, puissance de sortie et émissions hors bande sont conformes.
CEM (compatibilité électromagnétique)	EN 301 489-1 / EN 301 489-17	Le FCC Part 15 inclut les mesures de conduite et rayonnement (AC Conducted / Radiated Emission). Les résultats montrent des marges > 7 dB sous les limites. Les essais équivalents à CISPR 32 sont reconnus.
Exposition CEM globale (EMF)	EN 62311:2020 / IEC 62311:2020	Évaluation de champ rayonné équivalente à FCC §1.1310 / OET Bulletin 65. Exposition mesurée à 20 cm < 0,1 W/kg, bien en-dessous des limites européennes (1999/519/CE).

Exposition RF (SAR)	EN 62479 / EN 50663	Le module est prévu pour un usage fixe à distance ≥ 20 cm, donc exempt de mesure SAR directe. FCC §2.1091 & KDB 447498 fournissent une équivalence reconnue par la CE pour ce type d'appareil.
Bande autorisée (EU)	Décision CEPT 70-03 / EFIS	La bande 2400–2483,5 MHz est harmonisée SRD/Wi-Fi en Europe (100 mW EIRP). Le rapport FCC montre des EIRP ≤ 31 dBm avec antenne < 6 dBi – conforme si puissance ajustée à 20 dBm max.
Cybersécurité	ETSI EN 303 645 V2.1.1 (2020-06)	Non traité par FCC. Les modules Espressif intègrent WPA2-PSK, TLS et OTA sécurisés. Conforme aux recommandations IoT européennes.

The correspondence analysis between FCC tests and CE requirements demonstrates that the ESP32-C6-MINI-1 module fully satisfies the technical criteria of the RED Directive 2014/53/EU with respect to radio-frequency performance and electromagnetic compatibility.

The FCC tests conducted in accordance with ANSI C63.10-2013 and FCC Part 15 Subpart C §15.247 cover all RF parameters required by EN 300 328 V2.2.2, including output power, occupied bandwidth, out-of-band emissions, and frequency stability.

Similarly, the radiated and conducted emissions measurements are consistent with the limits defined in EN 301 489-1 and EN 301 489-17, ensuring the electromagnetic compatibility of the device.

CE compliance must be complemented by an electrical safety evaluation according to EN 62368-1 (which is not covered by the FCC report) and an RF exposure assessment under EN 62479, for which the module is exempt from SAR testing due to its low transmit power (< 20 dBm EIRP) and its use in a fixed installation at distances greater than 20 cm.

In conclusion, based on the available results, the ESP32-C6-MINI-1 radio module used in the Motion CAM-RF system can be considered technically compliant with the essential requirements of the RED Directive 2014/53/EU, provided that it is correctly integrated into a final product operating under the same installation conditions and power levels.

4.2 RF Filters and Spectral Compliance Masks

The ESP32-C6-MINI-1 module integrates a complete RF front-end that includes:

- a band-pass filter (BPF) centered at 2.45 GHz,
- a low-pass output filter on the transmit stage,
- a 50-ohm antenna matching network.

These filters are built directly into the RF signal chain in order to:

- suppress second- and third-order harmonics generated by the power amplifier
- ensure compliance with the spectral mask defined by EN 300 328 V2.2.2
- limit spurious emissions outside the ISM band (2400–2483.5 MHz)

- guarantee electromagnetic compatibility (EMC) with nearby equipment.

The internal filtering is complemented by the PCB antenna matching network (microstrip traces and tuning capacitors), optimized to maintain a VSWR $\leq 2:1$, which minimizes reflections and ensures stable transmission efficiency.

4.2.1 Compliance Masks

Spectral compliance is verified using the emission masks required by ETSI EN 300 328 and FCC Part 15.247.

These masks define the maximum allowable emission levels as a function of the frequency offset relative to the main carrier.

- Power and spectral density mask (ETSI EN 300 328, § 5.3.3) Largeur de bande d'émission (20 MHz) :
 - o Occupied bandwidth (20 MHz): compliant with the standard Wi-Fi channel width (IEEE 802.11 b/g/n).
 - o Maximum power spectral density: ≤ 10 dBm/MHz (European limit).
 - o Total EIRP: ≤ 20 dBm (100 mW).
- Out-of-band emission mask
 - o Emissions at ± 1 MHz from the band: ≤ -20 dBc.
 - o Emissions at ± 5 MHz and beyond: ≤ -30 dBm (standard limit).
 - o Harmonics (2nd and 3rd order): ≤ -36 dBm (as measured in the FCC report).

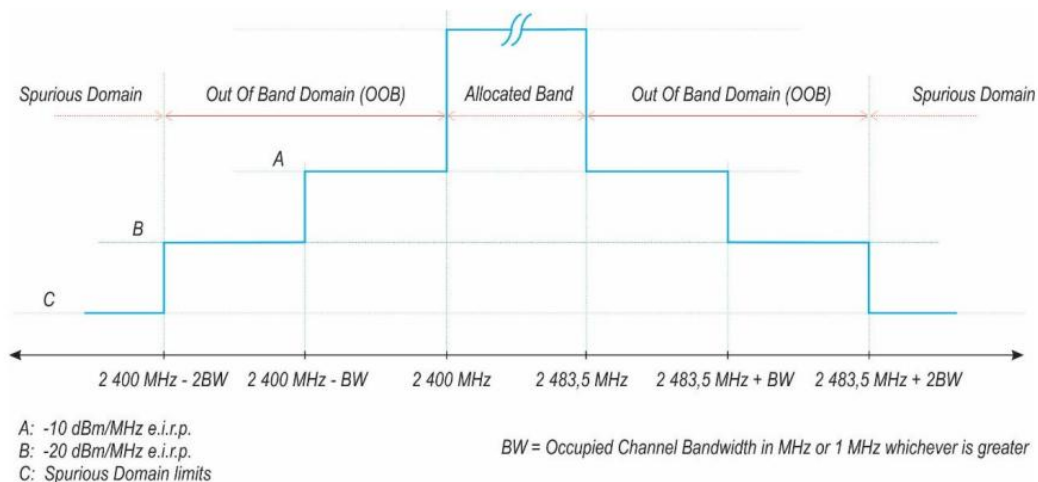


Figure 1: Transmit mask

Table 4: Transmitter limits for spurious emissions

Frequency range	Maximum power	Bandwidth
30 MHz to 47 MHz	-36 dBm	100 kHz
47 MHz to 74 MHz	-54 dBm	100 kHz
74 MHz to 87,5 MHz	-36 dBm	100 kHz
87,5 MHz to 118 MHz	-54 dBm	100 kHz
118 MHz to 174 MHz	-36 dBm	100 kHz
174 MHz to 230 MHz	-54 dBm	100 kHz
230 MHz to 470 MHz	-36 dBm	100 kHz
470 MHz to 694 MHz	-54 dBm	100 kHz
694 MHz to 1 GHz	-36 dBm	100 kHz
1 GHz to 12,75 GHz	-30 dBm	1 MHz

These measurements ensure that the transmitted signal complies with the Power Spectral Density (PSD) mask and does not interfere with other radio services.

- Immunity mask (EN 301 489-1)

For the reception path, the module must tolerate an incident field strength of 3 V/m (typical Wi-Fi level) without service degradation.

The receiver's input filtering (LNA + BPF) rejects out-of-band signals and helps prevent desensitization.

4.2.2 Interpretation and Compliance

The tests performed in the FCC report show out-of-band emissions below -36 dBm, which is stricter than the limits defined in EN 300 328.

The internal filtering of the ESP32-C6-MINI-1 module therefore ensures:

- compliance with the European spectral mask,
- control of spurious emissions in adjacent bands,
- sufficient electromagnetic compatibility for a residential indoor environment.

No external correction (such as an additional filter) is required, provided that the integration into the final product (Motion CAM-RF) meets the following conditions:

- maintaining a 50-ohm impedance between the module and the antenna,
- absence of any external amplification.

The internal filters of the ESP32-C6-MINI-1 module, together with the spectral compliance demonstrated in the FCC test report, ensure conformity with the requirements of RED 2014/53/EU, EN 300 328, and EN 301 489-1.

The Motion CAM-RF system can therefore be considered compliant with spectral purity and filtering requirements, without the need for any external filtering components, for a maximum transmission power of 20 dBm EIRP in the 2.4 GHz band.

The results show that the ESP32-C6-MINI-1 module meets the spectral purity, output power, and RF immunity requirements defined by the RED Directive 2014/53/EU and the associated harmonised standards. No corrective measures are required.

5 Technical Solution

This section describes the physical components of the system, their power supply, their role, and their key characteristics.

5.1 HC-SR501

The HC-SR501 is a passive infrared (PIR) sensor used to detect human motion based on variations in infrared radiation within its field of view. It is built around the LHI778 sensing element and the BISS0001 control IC, which handle signal processing and generate the corresponding logic output trigger.



Capteur de mouvement HCSR501

It detects variations in infrared radiation emitted by warm bodies (especially humans) moving within its optical field. When a significant change is detected, the module outputs a high TTL signal (3.3 V). This signal can then be read by a microcontroller—here, the M5Stack Tab5—to trigger an action such as displaying a video stream.

Main technical characteristics:

- Supply voltage: 5 to 20 V DC
- Current consumption: approximately 65 mA
- Output level: 3.3 V (high) / 0 V (low)
- Detection field: approximately 110 to 120°, with a range of 3 to 7 meters, adjustable via potentiometer
- Hold time (output active): adjustable from 5 seconds to 5 minutes
- Retrigger lockout: approximately 3 seconds after each detection
- Trigger modes:
 - o Single trigger (L): the output signal remains high for the programmed duration after the first detection.
 - o Repeat trigger (H): each new movement detected extends the duration of the high signal.
- Operating temperature: -15 °C to +70 °C
- Dimensions: 32 × 23 mm, Fresnel lens diameter 23 mm.

In the Motion CAM-RF system, the HC-SR501 is directly connected to a digital input on the M5Stack Tab5, which acts as the central controller. When motion is detected, the sensor generates a high logic signal that is transmitted to the Tab5, triggering the display of the live video stream from the ESP32-CAM over the 2.4 GHz Wi-Fi network.

The sensor's delay and sensitivity settings are adjusted to avoid false triggers while ensuring fast responsiveness (reaction time below 500 ms). This component offers the advantage of very simple integration (three wires: VCC, GND, OUT) and low power consumption, making it well suited for embedded or autonomous applications.

5.2 M5Stack Tab5

The M5Stack Tab5 is a multifunction IoT terminal based on a RISC-V architecture, serving as the central unit of the Motion CAM-RF system. It combines processing power, a touchscreen user interface, and wireless connectivity to manage the entire device: motion detection, video display, image recording, and network supervision.



M5stack Tab5

The module serves as the main control unit of the system. It receives the signal from the PIR sensor, establishes the Wi-Fi connection with the ESP32-CAM, displays the live MJPEG stream, and manages the display timing sequences. Thanks to its ESP32-P4 processor and its ESP32-C6-MINI-1U radio sub-module, it ensures smooth communication over the 2.4 GHz band while maintaining secure data exchanges (WPA2).

Main technical characteristics:

- Processor: ESP32-P4 (dual-core RISC-V 400 MHz + LP core 40 MHz)
- Wireless module: ESP32-C6-MINI-1U (Wi-Fi 6 / 2.4 GHz, internal PCB antenna + external MMCX connector)
- Memory: 16 MB Flash + 32 MB PSRAM
- Touchscreen: 5" IPS LCD – 1280×720 px, GT911 touch controller
- Integrated camera: 2 MP module (SC2356, 1600×1200)
- External storage: microSD slot up to 32 GB, FAT32 format
- Interfaces: USB-C OTG, USB-A Host, RS-485, M5-Bus, GPIO, MMCX antenna port
- Power supply: 5 V via USB-C or optional NP-F550 battery
- Dimensions: 128 × 80 × 12 mm
- Weight: approx. 125 g
- Operating temperature: 0 °C to +40 °C

The integrated microSD reader of the M5Stack Tab5 enables local storage of the images captured by the camera. In the Motion CAM-RF project, this feature is used to:

- automatically capture and save an image when motion is detected;
- store JPEG files locally for offline viewing;
- temporarily archive detection events in case of network connection loss.

Access to the SD card is handled through the Tab5's internal SPI bus, managed by the standard M5Unified / Arduino ESP32 API, which simplifies file handling and application development. The stored images can then be viewed directly through the touchscreen interface or transferred to a computer via the USB-C port.

Within the project architecture, the M5Stack Tab5 performs the following roles:

- Reception of the detection signal from the PIR sensor (HC-SR501);
- Automatic Wi-Fi connection to the smartphone's 2.4 GHz hotspot;
- Live display of the MJPEG stream from the ESP32-CAM with a latency under 5 seconds;
- Image recording on the microSD card upon each detection, with timestamping;
- Display timing management: the video stream remains active for 30 seconds after the last detection;
- Automatic reconnection: up to 5 attempts spaced 5 seconds apart in case of network loss;
- Error handling: 30-second timeout if the video stream becomes unreachable.

The program developed for the project (available in the project's GitHub repository) performs the following functions:

- It initializes the M5Stack Tab5 (display, GPIO, Wi-Fi, and microSD over SPI), then connects as a station to the configured hotspot (WPA2).
- A digital input (GPIO4: motion detector) serves as the trigger: when it goes high (PIR detection), the Tab5 periodically queries the ESP32-CAM via HTTP (GET `http://<IP>:<PORT>/capture`) to retrieve a JPEG image.
- The `fetchJpeg()` function robustly handles image reception, whether the response includes a Content-Length header (known size) or a chunked stream (unknown size), while maintaining controlled connection and read timeouts.
- Each received image is displayed in full screen; as long as the trigger remains active, a JPEG snapshot is saved to the microSD card every 3 seconds in the `/captures/` directory, with a unique filename generated from `millis()`.
- In the absence of detection, a standby interface is displayed once to avoid unnecessary refresh. The display refresh rate is limited by `kMinIntervalMs` (80 ms), and the main loop is kept lightweight (`delay(2)`) to ensure moderate CPU usage.

In summary: Wi-Fi connection → PIR wait state → JPEG acquisition from the ESP32-CAM → near real-time display → periodic microSD storage for as long as motion persists.

5.3 ESP32-CAM

The ESP32-CAM serves as the video capture and streaming module of the Motion CAM-RF system. Compact and energy-efficient, it integrates both a Wi-Fi/Bluetooth microcontroller and a digital image sensor, enabling the generation and transmission of a real-time video stream over a local network.

In this project, it provides the MJPEG stream that the M5Stack Tab5 retrieves and displays as soon as motion is detected.



Esp32-cam

The ESP32-CAM is based on the ESP32-S microcontroller from Espressif Systems, featuring a dual-core Tensilica LX6 processor running at up to 240 MHz. It handles 2.4 GHz Wi-Fi connectivity (802.11 b/g/n) and Bluetooth 4.2 while performing image capture and processing from the integrated OV2640 sensor.

The module includes an embedded HTTP server that streams compressed MJPEG video, accessible from any web browser or any device connected to the same local network. This operating mode makes the camera fully autonomous, requiring no intermediary server.

Main technical characteristics:

- Processor: ESP32-S (dual-core LX6 @ 240 MHz)
- Memory: 520 kB SRAM + 4 MB external PSRAM
- Camera: OV2640 – resolution: 2 MP (1600 × 1200 px), field of view: 65–70°
- Image format: JPEG / MJPEG
- Wireless connectivity: 2.4 GHz Wi-Fi (IEEE 802.11 b/g/n), Bluetooth 4.2 BR/EDR + BLE
- Supply voltage: 5 V DC (5 V pin or FTDI adapter)
- Typical power consumption: 160 to 310 mA depending on transmission mode
- Interfaces: UART, SPI, I²C, GPIO (flash LED control, external trigger, etc.)
- Dimensions: 27 × 40 mm
- Operating temperature: –20 °C to +70 °C

In this project, the ESP32-CAM operates as an autonomous video node:

- It is powered with 5 V through a USB supply or a computer.
- It connects to the smartphone's 2.4 GHz Wi-Fi hotspot, on the same network as the M5Stack Tab5.
- It streams an MJPEG video feed accessible at a dynamic IP address (e.g., <http://192.168.x.x:81/stream>).
- When motion is detected by the PIR sensor, the Tab5 establishes an HTTP connection to this address and immediately displays the video stream.
- In the event of a network outage, the camera automatically restarts its HTTP server once the access point becomes available again.
- This simplified operating mode ensures high responsiveness and an overall latency below 5 seconds, meeting the project's functional requirements.

The program developed for the project (available in the project's GitHub repository) provides the following functionalities:

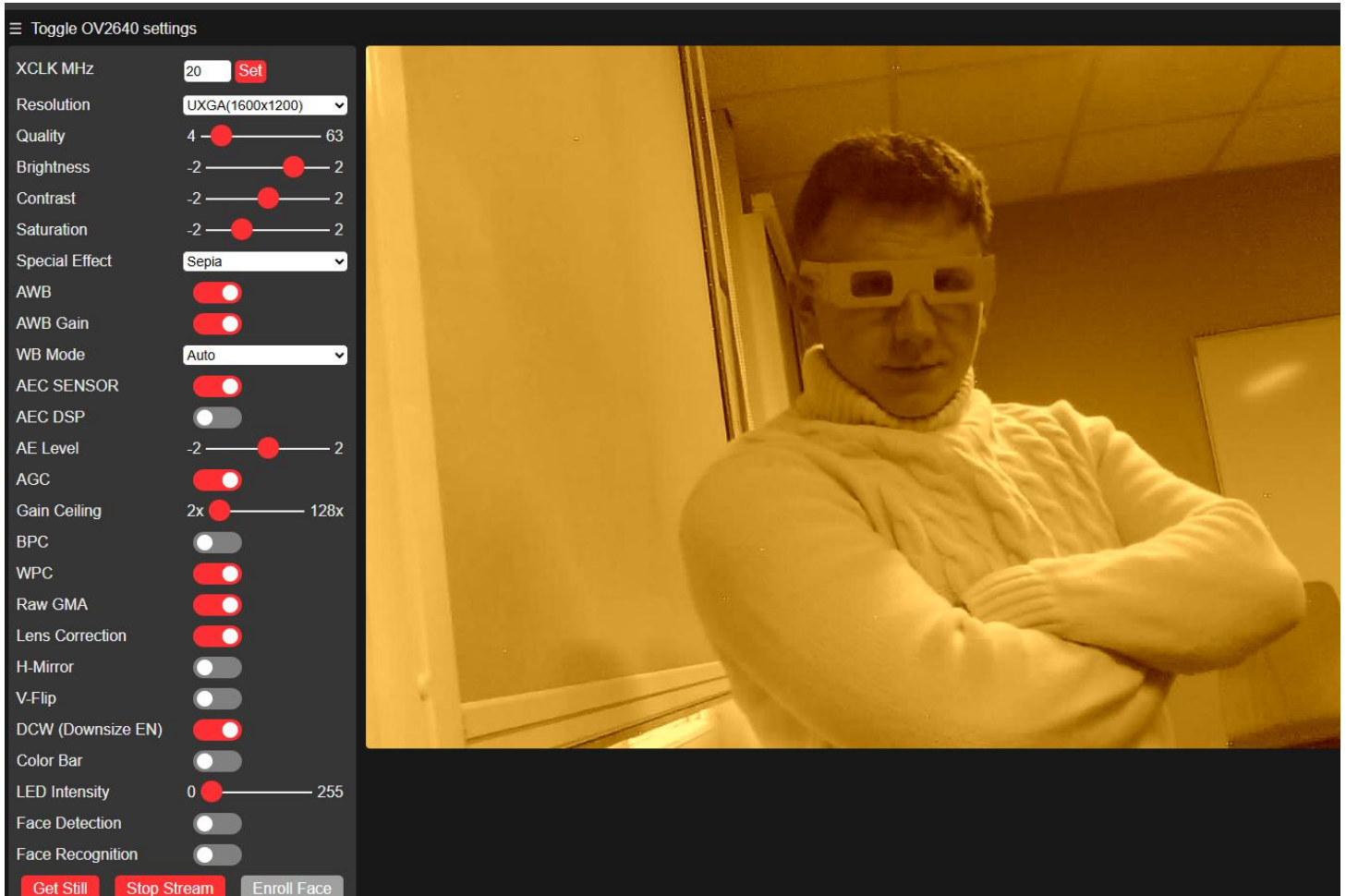
- It configures the OV2640 camera sensor integrated into the ESP32-CAM board, as well as the entire image-capture pipeline (parallel bus D0–D7, VSYNC/HREF/PCLK signals, and SCCB interface for I²C control).
- The XCLK clock is set to 20 MHz, the image format to PIXFORMAT_JPEG, and the frame buffers are allocated in PSRAM when available.

- If PSRAM is detected, the camera operates in HD resolution (1280×720) with high JPEG quality (quality factor 10) and two frame buffers for smooth streaming; otherwise, the code reduces the image size and uses a single buffer. Sensor parameters (mirror, vertical flip, brightness, saturation, quality) are adjusted for optimal rendering.
- Once the camera is initialized, the module connects in Wi-Fi station mode (WPA2) to the configured access point. Wi-Fi power-saving is disabled to ensure stable video streaming. The assigned local IP address is printed on the serial monitor.
- Two HTTP servers are then started:
 - /status: returns a JSON object containing the camera parameters (resolution, quality, brightness, etc.).
 - /control: allows dynamic modification of capture parameters (contrast, brightness, image size, orientation, etc.).
 - /capture: performs an instant JPEG capture accessible through a simple HTTP request.
 - /bmp: captures an image in BMP format.
 - Additional technical endpoints: /xclk, /pll, /reg, /greg, /resolution for fine-tuning the sensor configuration.
- Video streaming server:
 - /stream: provides a continuous MJPEG stream over HTTP (multipart/x-mixed-replace). Each frame includes its size and a timestamp. The code measures the time between frames and computes an average frame rate displayed in the logs.

The ESP32-CAM also includes a built-in web-based Human–Machine Interface (HMI), accessible through any web browser, which allows live image viewing and fine camera control without requiring additional tools. Once connected to the smartphone's Wi-Fi network, the camera exposes a control page at the local IP address displayed at startup (e.g., <http://192.168.x.x/>).

This HMI, provided natively within the firmware, gives access to several essential features: resolution adjustment, JPEG quality, brightness, contrast, LED flash control, selection of the capture mode (single image or MJPEG stream), image mirroring or rotation, and real-time viewing of the current frame rate.

It therefore offers a simple and intuitive way to test the camera, adjust optical settings according to lighting conditions, and validate the correct operation of the video stream independently of the M5Stack Tab5. This interface is a valuable tool for diagnostics, fine tuning, and maintenance of the Motion CAM-RF system.



IHM espcam

The image buffers are systematically released after transmission to preserve memory and prevent leaks. CORS headers are included to allow access from an external client, such as the M5Stack Tab5 tablet.

In summary, this program transforms the ESP32-CAM into a Wi-Fi video capture and streaming server. It provides a simple HTTP interface enabling either the retrieval of a single image (/capture) or access to a continuous MJPEG stream (/stream), while also allowing remote adjustment of imaging parameters and control of an auxiliary LED light.

6 Troubleshooting

This section lists the main issues that may be encountered during the integration or use of the Motion CAM-RF system, along with their probable causes and corresponding solutions. It is based both on difficulties observed during development and on the inherent hardware limitations of the components used.

6.1 MicroSD Card Not Recognized by the M5Stack Tab5

Symptom: The Tab5 does not display any media, saving fails, or the function `fs::FS::open()` returns an error.

Cause: The M5Stack Tab5 only supports microSD cards formatted in FAT32. Any card formatted in exFAT or NTFS will not be recognized by the M5Unified / ESP-IDF APIs.

Solution: Reformat the microSD card to FAT32 (32 kB allocation unit).

6.2 Unable to Use a USB Camera with the Tab5

Symptom: Connecting a USB webcam results in no detection, and it is not possible to open a video stream via USB-OTG.

Cause: The M5Stack Tab5 has no built-in UVC video driver and no Arduino-compatible library capable of handling a USB camera.

Although the ESP32-P4 supports USB-OTG, the software ecosystem does not provide any ready-to-use stack for video capture.

Solution: Use a network camera (ESP32-CAM) as implemented in this project, or develop a custom UVC driver from scratch based on ESP-IDF.

6.3 Frequent Wi-Fi Disconnections Between the Tab5 and the ESP32-CAM

Symptom: The MJPEG stream freezes, and the Tab5 can no longer reach `/stream` or `/capture`.

Possible causes:

- The smartphone's hotspot automatically changes its Wi-Fi channel.
- The distance is too great (>10 m) or obstacles (load-bearing walls) attenuate the signal.
- The ESP32-CAM is configured with `wifi_ps_mode(WIFI_PS_MIN_MODEM)`, causing micro-interruptions.
- Network congestion due to an MJPEG bitrate that is too high.

Solutions:

- Fix the smartphone's Wi-Fi channel (ideally channel 1, 6, or 11).
- Disable Wi-Fi power-saving mode on the camera.
- Reduce the MJPEG resolution (QVGA / VGA).

- Ensure the ESP32-CAM has a stable power supply (≥ 500 mA).

6.4 Overheating and Instability of the ESP32-CAM at High Resolution

Symptom: Unexpected reboots, Brownout detected messages, or a drastic drop in framerate.

Cause: The module draws significantly more current at high resolutions (SVGA, XGA, UXGA), especially when the flash LED is enabled. An unstable USB power supply can lead to resets.

Solutions:

- Use a dedicated 5 V / 1 A power supply.
- Avoid resolutions higher than 1280×720 unless a heatsink is used.
- Disable the flash LED unless strictly necessary.

6.5 Video Latency Greater Than 5 Seconds

Symptom: The video stream takes too long to appear on the Tab5 despite a seemingly good connection.

Possible causes:

- MJPEG compression set too high (low quality = larger frames).
- Buffering issues on the Tab5 during full-screen display.
- Unstable Wi-Fi throughput (overloaded hotspot or smartphone throttling).

Solutions:

- Set the JPEG quality on the ESP32-CAM between 8 and 12.
- Limit the capture framerate to 8–12 FPS.
- Temporarily disconnect other devices from the hotspot.
- Keep the camera away from interference sources (microwave ovens, 2.4 GHz routers).

7 Conclusion

The Motion CAM-RF project resulted in the development of an embedded system for motion detection and wireless video transmission that complies with European technical and regulatory requirements. The radio-frequency analysis confirmed the system's compatibility with the RED Directive 2014/53/EU and the ETSI EN 300 328 and EN 301 489-17 standards, ensuring safe and stable operation within the 2.4 GHz ISM band.

The entire system also meets safety requirements (EN 62368-1) and electromagnetic compatibility standards, guaranteeing reliable operation in an indoor environment.

This work demonstrates proficiency in RF engineering, regulatory compliance, and embedded systems design, while laying the groundwork for future laboratory validation and potential evolution toward a pre-industrial version.

8 Appendices

8.1 RADIO ELECTRONIC SYSTEM SPECIFICATIONS (claudio.ai)

PROJECT INFORMATION

- Project Name: MOTION CAM-RF
- Designer: Brice BENINI
- Date: 2025-10-31
- Version: 1.0

1. GENERAL SPECIFICATIONS

- System Type: Half-duplex bidirectional Wi-Fi transceiver
- Device Category: Short Range Device (SRD)
- CEPT Annex: Annex 1
- Station Type: Wi-Fi Station (End-user terminal)
- Application: Commercial
- Intended Use: Short-range Wi-Fi communication for local video transmission and motion detection monitoring
- Main Target Market: France
- Operating Mode: Duplex

2. FREQUENCY SPECIFICATIONS

- Frequency Range: 2400 MHz - 2483.5 MHz
- Center Frequency: 2 442 MHz
- Channel Spacing: 5 MHz
- Bandwidth: 20 MHz
- EFIS Verified: Yes
- EFIS Notes: Bande vérifiée dans la base EFIS pour usage Wi-Fi / ISM (2400-2483.5 MHz, 100 mW EIRP)

3. TRANSMITTER SPECIFICATIONS

- Output Power: 100mW
- Frequency Stability: 20ppm
- Spurious Emission: <-36 dBm
- Modulation: OFDM / OFDMA (Orthogonal Frequency Division Multiplexing / Multiple Access)

4. RECEIVER SPECIFICATIONS

- Sensitivity: -90 dBm
- Selectivity: >35 dB
- Noise Figure: ≤ 7 dB
- Dynamic Range: 80 dB

5. ANTENNA SPECIFICATIONS

- Implementation: PCB Printed Antenna
- Type: Monopôle
- Gain: 2 dBi
- VSWR: ≤ 2.0 : 1
- Polarization: Linéaire

6. FEATURES & CHARACTERISTICS

- Input Voltage: 5V
- Power Consumption: RX : 80 mA TX : 380mA
- Operating Temperature: 0 to 40°C
- Humidity: 60%RH
- Dimensions: 13.2 × 12.5 × 2.4 mm
- Weight: 0.2Kg

7. RF CHIPSET

- Manufacturer: Espressif
- Model: ESP32-C6-MINI-1

- Details:
- Application Note:

10. CYBERSECURITY

- Applicable Standards: ETSI EN 303 645

11. STANDARDS & CERTIFICATIONS

- Genorma Search: <https://genorma.com/en/project/advanced-search?keywords=Commercial&status%5B0%5D=PUBLISHED&status%5B2%5D=WITHDRAWN&operatorCommittee=1&operatorStdType=3&listMode=DEFAULT>
- Market Access (360compliance):
<https://360compliance.co/marketaccess/france/>
- IEC Safety: IEC 62368-1:2018
- IEC EMC: IEC 61000-6-1:2016 / IEC 61000-6-3:2020
- CENELEC Safety: EN 62368-1:2020
- CENELEC EMC: EN 301 489-1:2022 / EN 301 489-17:2022
- ETSI: EN 300 328 V2.2.2 (2019-07)
- IEEE: IEEE 802.11
- 3GPP: Not applicable (Wi-Fi system)
- Certifications: CE, FCC, IC, UKCA, RCM

12. SAR SAFETY

- Foreseeable Use: Not specified
- SAR Limit: Other
- Notes: usage sans contact humain direct, distance ≥ 20 cm, dispositif fixe, utilisation en intérieur, non porté.

13. USE CASES

- Use Case 1: Surveillance intérieure déclenchée par mouvement: Un système de surveillance intérieure détecte un mouvement grâce à un capteur infrarouge passif (PIR) et affiche automatiquement le flux vidéo d'une caméra ESP32-CAM via le réseau Wi-Fi local (2,4 GHz). La tablette M5Stack Tab5 sert d'interface d'affichage et de commande. Ce cas d'usage illustre une solution simple de vidéosurveillance locale pour un appartement, un bureau ou une salle de laboratoire.
- Use Case 2: Détection de présence domestique : Dans un environnement résidentiel, le système détecte la présence d'une personne dans une pièce (entrée, couloir, salon) et affiche en temps réel le flux vidéo associé sur l'écran de la Tab5. La communication entre les modules s'effectue sur un réseau Wi-Fi fermé (partage de connexion du smartphone). Le dispositif garantit une latence faible (< 5 s) et le respect de la confidentialité, puisque les données ne quittent pas le réseau local.
- Use Case 3:

14. FUNCTIONAL REQUIREMENTS

Requirement 1: Détection de mouvement

- Sub-requirement 1.1: Le capteur PIR (HC-SR501) doit détecter toute variation thermique correspondant à une présence humaine.
- Sub-requirement 1.2: Le signal de détection doit être transmis à la M5Stack Tab5 via une entrée GPIO, avec un anti-rebond logiciel de 500 ms.

Requirement 2: Affichage du flux vidéo

- Sub-requirement 2.1: Lorsqu'un mouvement est détecté, la Tab5 doit afficher automatiquement le flux vidéo MJPEG issu de la caméra ESP32-CAM.
- Sub-requirement 2.2: Le flux doit apparaître en plein écran, avec une latence maximale de 5 s entre la détection et l'affichage.

Requirement 3: Communication Wi-Fi

- Sub-requirement 3.1: Le système doit utiliser la bande 2,4 GHz (Wi-Fi IEEE 802.11 b/g/n) via un point d'accès local (hotspot smartphone).
- Sub-requirement 3.2: Le protocole de sécurité doit être WPA2-PSK, et la connexion doit se rétablir automatiquement après une perte de signal.